



Calculating theoretical probabilities from a table (one event)

Task A: Finding probabilities from a frequency table

1) The table shows the number of sweets of each flavour in a novelty box of sweets.

flavour	frequency
strawberry	11
caramel	10
toffee	8
fudge	12
orange	8
chocolate	9
grass	8
earwax	3
fish	5
garlic	6
total	80

a) Find the probabilities for picking each of the following flavours when a sweet is chosen at random.

$$P(\text{strawberry}) = \frac{11}{80}$$

$$P(\text{caramel}) = \frac{10}{80}$$

$$P(\text{toffee}) = \frac{8}{80}$$

$$P(\text{chocolate}) = \frac{9}{80}$$

$$P(\text{grass}) = \frac{8}{80}$$

$$P(\text{earwax}) = \frac{3}{80}$$

$$P(\text{fish}) = \frac{5}{80}$$

$$P(\text{garlic}) = \frac{6}{80}$$

b) The nasty flavours are grass, earwax, fish and garlic. What is the probability of choosing a nasty flavoured sweet?

$$\frac{22}{80}$$

2) Sam is carrying out a data investigation about how pupils in a nearby village travel to school. She conducts a survey and records her results in a tally chart.

a) Complete the frequency column of the table.

mode of transport	tally	frequency
car		20
bus		12
bicycle		9
walk		16

b) To further her investigation, Sam chooses a pupil at random to interview.

i) What is the probability that the person chosen travels by car? $\frac{20}{57}$

ii) What is the probability that the person chosen travels by bus? $\frac{12}{57}$

iii) What is the probability that the person chosen travels by bicycle? $\frac{9}{57}$

iv) What is the probability that the person chosen walks to school? $\frac{16}{57}$

Task B: Finding probabilities from an outcome table

1) Party hats from a pack are sorted according to how they are decorated.

A hat is chosen at random from the pack.

Find each of the following probabilities.

a) $P(\text{stripes and topper}) = \frac{2}{10}$

b) $P(\text{stripes and no topper}) = \frac{3}{10}$

c) $P(\text{no stripes and topper}) = \frac{4}{10}$

d) $P(\text{no stripes and no topper}) = \frac{1}{10}$

e) $P(\text{stripes}) = \frac{5}{10}$

f) $P(\text{topper}) = \frac{6}{10}$

	stripes	no stripes
topper		
no topper		

2) A deck of cards contains the numbers 1 to 20.

a) Sort the numbers 1 to 20 into the table.

b) A card is chosen at random from the deck.

Find the probabilities of the following outcomes.

i) $P(\text{multiple of 4 and factor of 20}) = \frac{2}{20}$

ii) $P(\text{multiple of 4 and not factor of 20}) = \frac{3}{20}$

iii) $P(\text{multiple of 4}) = \frac{5}{20}$

iv) $P(\text{not multiple of 4}) = \frac{15}{20}$

	multiple of 4	not multiple of 4
factor of 20	4, 20	1, 2, 5, 10
not factor of 20	8, 12, 16	3, 6, 7, 9, 11, 13, 14, 15, 17, 18, 19

